Thiago de Araujo Silva

Technical Artist / Junior Tools & UI Programmer

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PROFILE

Technical Artist with a background in game programming (**VFS**) and technical arts (**BCIT**). I work at the intersection of artists, designers, and engineers, building tools, pipelines, and procedural setups in Houdini so teams can get assets into Unreal and Unity faster and with fewer headaches.

SKILLS

Languages & Shaders:	Python, C#, VEX, HLSL, GLSL, ShaderLab, JavaScript, C++
3D & VFX:	Houdini (Procedural Content, Smoke, Explosions, Fractures, Cloth), Maya, Blender
Engines & Tools:	Unreal Engine, Unity, Perforce, Git, Photoshop, Illustrator, Krita, GIMP

EDUCATION & TRAINING

Advanced Diploma, Technical Arts — **BCIT** 2022 – 2023

- Focused on tools, DCC-to-engine pipelines, procedural content generation in Houdini (VEX), and real-time VFX for Unreal Engine and Unity.

Programming for Games, Web & Mobile — **Vancouver Film School (VFS)** 2015 – 2016

- Trained in C++, C#, game architecture, math for games, and shaders. Won four awards: *Excellence in Architecture*, *Most Innovative Project (Vulkan)*, and two *Best Final Game* awards.

Game Dev Program — **PUC Minas (Brazil)** | **Drawing & 3D Modeling** — **Casa Dos Quadrinhos** 2009 – 2014

TECHNICAL PROJECTS & PIPELINE TOOLS

Retro LUT Compiler — **Python, Custom Shaders (HLSL/GLSL)**

- Developed a standalone Python tool to generate discrete Look-Up Table (LUT) files mimicking retro game console color palettes.
- Designed and implemented both ends of the pipeline: the custom shader consuming the palette data and the compiler tool generating the LUT assets.

Isometric Sprite Pipeline — **Houdini, Python, HDA, TOPs**

- Built an automated batch-rendering pipeline in Houdini utilizing TOPs to convert 3D assets into isometric sprite sheets with normal maps and cutout alpha textures.
- Engineered custom HDAs for dynamic camera framing, optimal power-of-2 cropping, and automated sprite-sheet assembly.

World Machine & Unreal Terrain Pipeline — **World Machine, Unreal Engine**

- Generated heightmaps and splat maps in World Machine, integrating them into Unreal landscapes with custom shader optimization.
- Packed albedo luminosity maps into ORD texture alpha channels, driven by a massive color map to break texture repetition and bypass sampler limits.

AI-Gen to Procedural Diorama & 2.5D Scenes — **AI Art, Unity**

- Created an interactive main menu by slicing AI-generated landscape artwork into distinct depth panels and assembling a layered 3D diorama.
- Implemented smooth UI state transitions, camera panning, and animated unfolding sequences for submenu navigation.

Adventurers Needed (VFS Final Project) — **Software Architecture, C#, FX Art**

- Designed a layered C# system in Unity to coordinate multiple programmers, built an asset pipeline, and created custom VFX and rendering setups.